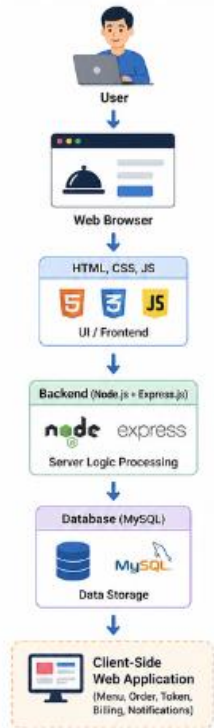
College Canteen Token System

Project Based Learning From FEDF

Abstract:

The College Canteen Token System is a web-based application developed to improve the efficiency and convenience of food ordering in college canteens. Traditional manual ordering methods often lead to long queues, delays, confusion in token management, and increased workload for canteen staff, especially during peak hours. The proposed system aims to address these issues by providing a digital platform where students can browse menu items, apply dietary filters, place food orders online, and receive a unique token for organized food collection.

System Architecture



CO - WISE IMPLEMENTATION

CO 1- Foundations of Front-End Engineering & Framework Design

- React components: Home, About, Contact, Login
- Navigation using React Router
- Responsive UI with CSS

```
import Home from "./Home";
import About from "./AboutUs";
import Contact from "./ContactUs";
import Login from "./Login";
```

CO 2- JavaScript & TypeScript Engineering for Frameworks

- useState: Login inputs, Cart system
- Event handling: Button clicks (Add to cart, Place Order), Input handling

```
const [cart, setCart] = useState([]);

const addToCart = (item) => {
  setCart([...cart, item]);
};
```

CO 3 – React Component Model

- Menu array (data structure)
- Dynamic rendering using .map()
- Cart logic (adding items)

```
<div className="features">
  {menuItems.map((item) => (
    <div className="card" key={item.id}>
```

CO 4 – State Architecture, Async Data Engineering

- Simulated real-world system:
- Digital ordering
- QR-based payment

```
<button
  style={{ marginTop: "10px" }}
  onClick={() => navigate("/payment")}
>Place Order</button>
```

CO5 – Routing, Forms, Accessibility & Performance Engineering

- Total calculation
- Data passing (token + amount)

```
const total = cart.reduce((sum, item)
 => sum + item.price, 0);
```

CO 6- Build Systems, Testing, CI/CD & Deployment Engineering

- Token-based system design
- Real-world workflow simulation

```
const token = Math.floor(100 + Math.random() * 90);
```

Conclusion and Future Scope:

This project successfully implements a digital college canteen system with menu display, order placement, token generation, and QR-based payment simulation, reducing waiting time and improving efficiency. In the future, it can be enhanced by integrating a backend database, real payment gateways, admin dashboard, and live order tracking to make it a fully functional real-world application.

Cluster: 1
Section: 2
Team Members
2520080007 B. Harshitha
2520090007 S. Dikshita